LESSON 05 · MISSION GUIDE

# The Export

Strings, files and CSVs: getting data in and out of Python without opening Excel.

From **Priya, Payroll**
To **Jen**
Subject **The export (again)**

Jen, it's the first of the month, so the HR system has produced The Export. Forty rows. Refs in three styles, names in CAPITALS or whispering in lowercase, salaries with or without pound signs, hours with "hrs" stuck on the end. Every month I fix it by hand in Excel, and every month Excel turns the start dates American.

Mo says Python can read the file as it is, clean it, write a new one, and do exactly the same next month. I'd like that before Thursday, because Gerald told the board that Production is our most expensive department and now somebody has to check. — P.

## What you'll be able to do after today

- [ ] Take strings apart with `.split()` and put them back together with `.join()`
- [ ] Write text to a file and read it back, line by line or all at once
- [ ] Read a CSV with the `csv` module, as a list of dictionaries, one per row
- [ ] Clean a messy export with the string tools from lesson 01, in a loop, with functions
- [ ] Write a clean CSV that Excel, Siobhan and next month's script can all open
- [ ] Find and describe files with `pathlib`

### Why this matters at work

- **Every analysis starts with a file.** Exports, extracts, downloads: they arrive as text, and somebody has to make sense of them. From today, that somebody is a script, and it never gets bored on row 37.
- **Excel changes your data when it opens it.** Leading zeros vanish, "01/02" becomes a date, long numbers become 1.2E+15. Python reads exactly what's there and nothing else.
- **The clean file is the product.** Once the export is clean, every later lesson (and the pay gap report) starts from it.

## Four big ideas

| | |
|---|---|
| **1. `with open(...) as f`** | Opens a file and closes it again automatically, even if something goes wrong in between. Always use it. |
| **2. A file is just lines of text** | Even a CSV. Each line ends in `"\n"`, and every value is text until you convert it. |
| **3. The `csv` module does the fiddly bits** | Commas inside quotes ("Robinson, Max"), the header row, writing it all back out. Don't `.split(",")` a CSV yourself. |
| **4. Paths are objects** | `Path("data") / "payroll.csv"` builds a path that works on Windows and Mac, and can list, read and check files. |

### Three things that bite on Windows

- Open CSVs with `newline=""`, or you get a blank line after every row you write.
- Add `encoding="utf-8"` whenever a file might contain £ or a name with an accent. Without it, Windows guesses, and guesses wrong.
- Paths print with backslashes (`my_work\report.txt`). Use forward slashes in your code; Python translates.

## How today works

15 MIN
**Lecture**
Lars opens the export live and reads it three ways

60 MIN
**Exercise sheet**
Seven tickets, from strings to a finished clean file

15 MIN
**Dave's Leftovers #5**
His stats script. Three bugs. One typo

## The inbox

| # | FROM | TICKET | YOU'LL USE |
|---|---|---|---|
| 1 | Mo | Split and join | `.split() .join() set() sorted()` |
| 2 | Gerald | Write a file, read it back | `open("w") .write() .read() for line in f` |
| 3 | Priya | Open the export | `csv.DictReader fieldnames` |
| 4 | Priya | Clean it up | functions + `.strip() .title() .replace()` |
| 5 | Siobhan | Write the clean file | `csv.DictWriter writeheader() writerows()` |
| 6 | Gerald | Department totals | dicts + `.get(key, 0)` + `max(key=)` |
| 7 ⭐ | Mo | Files about files | `Path .glob() .read_text()` |

## Your three documents

| | |
|---|---|
| **GUIDE.pdf** | This one. |
| **EXERCISES.pdf** | The exercise sheet, with the expected output to check against. |
| **CHEATSHEET.pdf** | Strings, files, csv and pathlib on one page. Keep it open. |

### You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if you're brave), each printing the expected output
- [ ] Dave's script prints the four expected lines, one commit per bug
- [ ] Everything is pushed to `lesson-05` and you've opened a pull request




**Next lesson: Monday Morning.** The export has a row that says "n/a" where a salary should be, and your script falls over on it. You'll learn to read tracebacks properly, catch errors on purpose, and make scripts that survive real data.